FIELD-VALUED MONOMORPHISM POINTS OF A DECENT SPACE CLASSIFY ITS POINTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2577-2585, immediately following the statement at 2564-2575.

Editorial note. The parenthetical '(actually one can say this is part of the definition of being decent)' at source lines 2582-2583 is retained verbatim; it marks the surjectivity half as a definitional reading.

Editorial note. One cross-chapter pointer (Properties of Spaces) and one same-chapter line-indexed stub are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Injectivity of the monomorphism-point map. Properties of Spaces, Lemma spaces-properties-lemma-points-monomorphism, cited at proof lines 2577-2579. Carries the injectivity half in general (no decency needed).
- Surjectivity as part of decency. Lemma lemma-bounded-fibres of this chapter, cited at proof lines 2580-2583 (with the author's remark that this is part of the definition of being decent). Carries the surjectivity half when X is decent.

Source reference. Original lemma 03JX: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 952–952. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03E1: focused-seeds/geometry/S9-decent-decent-points-monomorphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Consider the map*

$$\{\mathrm{Spec}(k) \to X \text{ monomorphism where } k \text{ is a field}\} \longrightarrow |X|$$

This map is always injective. If X is decent then this map is a bijection.

Proof. We have seen in Properties of Spaces, Lemma 03E1 that the map is an injection in general. By Lemma 03JX it is surjective when X is decent (actually one can say this is part of the definition of being decent). □

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